

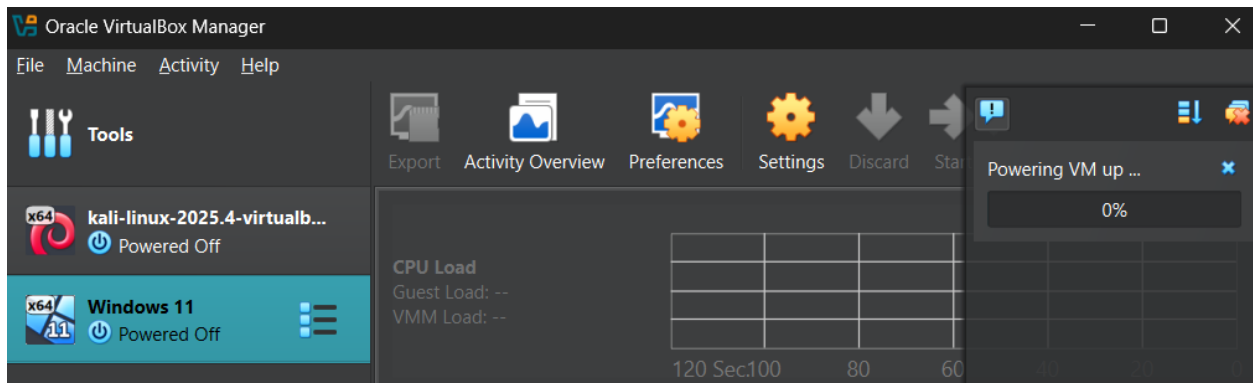
Activity Documentation

Title: Creating a Bash Script to Generate a Directory Structure

Objective

To create and execute a Bash script that automatically generates the required directory and file structure in the current working directory.

Step 1. Start the Virtual Machine



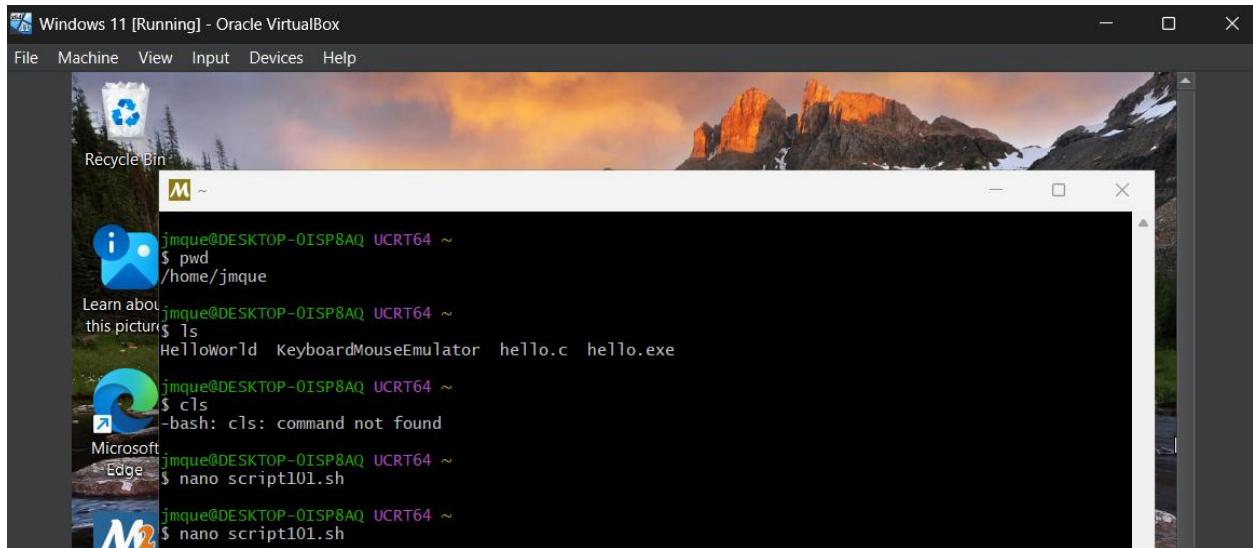
Step 2. Open the Terminal

```
M ~
jmqque@DESKTOP-0ISP8AQ UCRT64 ~
$ pwd
/home/jmqque
jmqque@DESKTOP-0ISP8AQ UCRT64 ~
$ ls
HelloWorld KeyboardMouseEmulator hello.c hello.exe
jmqque@DESKTOP-0ISP8AQ UCRT64 ~
$ |
```

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Step 4. Create the Bash Script



```
jmq@DESKTOP-0ISP8AQ UCRT64 ~  
$ pwd  
/home/jmq  
jmq@DESKTOP-0ISP8AQ UCRT64 ~  
$ ls  
HelloWorld KeyboardMouseEmulator hello.c hello.exe  
jmq@DESKTOP-0ISP8AQ UCRT64 ~  
$ cls  
-bash: cls: command not found  
jmq@DESKTOP-0ISP8AQ UCRT64 ~  
$ nano script101.sh  
jmq@DESKTOP-0ISP8AQ UCRT64 ~  
$ nano script101.sh
```



```
GNU nano 9.0 script101.sh  
  
[ New File ]  
Help Exit Write Out Read File Where Is Replace Cut Paste New File Execute Justify Location Go To Line Undo Redo
```

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Step 4. Type the Script

```
#!/usr/bin/env bash

mkdir -p OS/P1
mkdir -p OS/P4
mkdir -p OS/Network/P2
mkdir -p OS/Network/P3

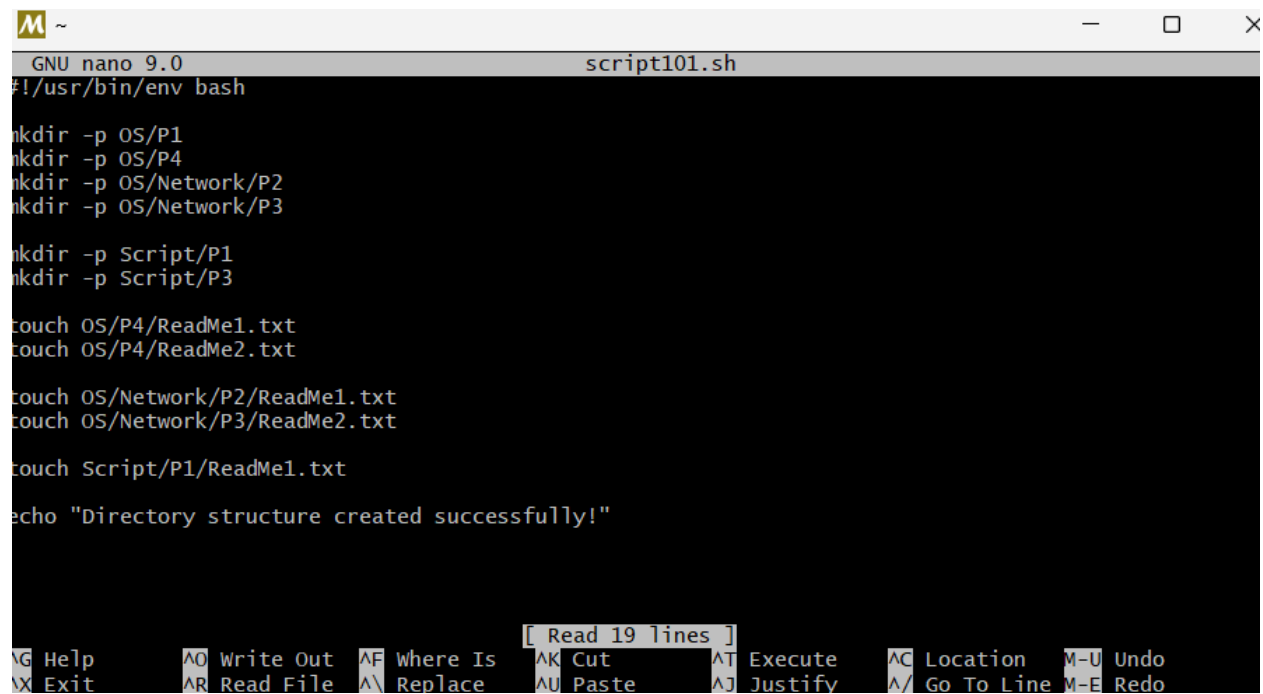
mkdir -p Script/P1
mkdir -p Script/P3

touch OS/P4/ReadMe1.txt
touch OS/P4/ReadMe2.txt

touch OS/Network/P2/ReadMe1.txt
touch OS/Network/P3/ReadMe2.txt

touch Script/P1/ReadMe1.txt

echo "Directory structure created successfully!"
```

A screenshot of a terminal window titled "script101.sh" running GNU nano 9.0. The terminal displays the same script content as the previous block. The bottom status bar shows various nano editor shortcuts: ^G Help, ^O Write Out, ^F Where Is, ^K Cut, ^T Execute, ^C Location, ^M-U Undo, ^X Exit, ^R Read File, ^N Replace, ^U Paste, ^J Justify, ^_ Go To Line, and ^M-E Redo. A message "[Read 19 lines]" is also visible in the status bar.

```
GNU nano 9.0 script101.sh
#!/usr/bin/env bash

mkdir -p OS/P1
mkdir -p OS/P4
mkdir -p OS/Network/P2
mkdir -p OS/Network/P3

mkdir -p Script/P1
mkdir -p Script/P3

touch OS/P4/ReadMe1.txt
touch OS/P4/ReadMe2.txt

touch OS/Network/P2/ReadMe1.txt
touch OS/Network/P3/ReadMe2.txt

touch Script/P1/ReadMe1.txt

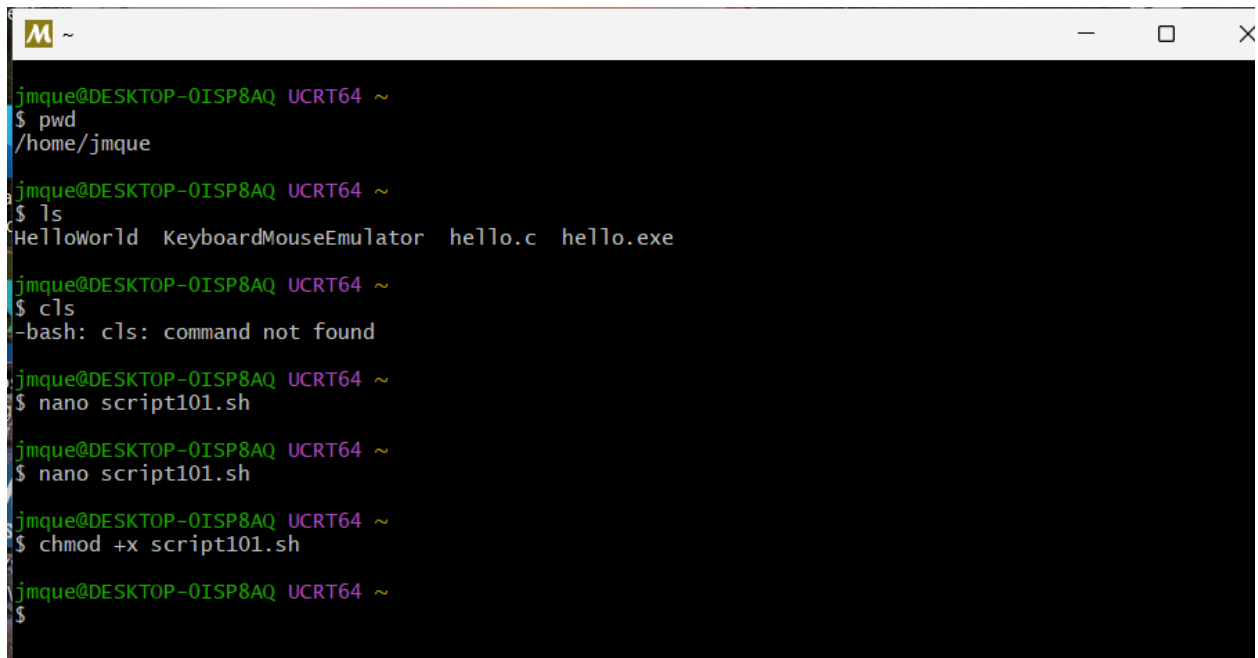
echo "Directory structure created successfully!"

[ Read 19 lines ]
^G Help  ^O Write Out  ^F Where Is  ^K Cut  ^T Execute  ^C Location  ^M-U Undo
^X Exit  ^R Read File  ^N Replace  ^U Paste  ^J Justify  ^_ Go To Line  ^M-E Redo
```

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Step 6. Give Execute Permission

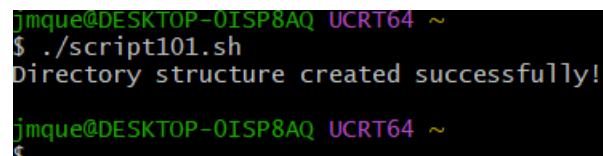
`chmod +x script101.sh`



```
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ pwd  
/home/jmque  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ ls  
HelloWorld KeyboardMouseEmulator hello.c hello.exe  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ cls  
-bash: cls: command not found  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ nano script101.sh  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ nano script101.sh  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ chmod +x script101.sh  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$
```

Step 7. Execute the Script

`./script101.sh`

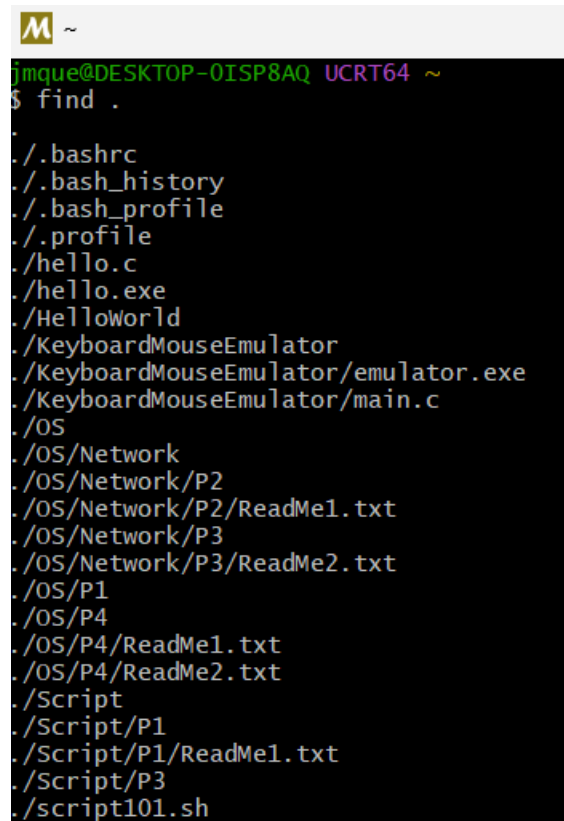


```
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$ ./script101.sh  
Directory structure created successfully!  
  
jmque@DESKTOP-0ISP8AQ UCRT64 ~  
$
```

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Step 8. Verify the Directory Structure

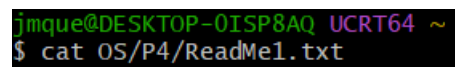
find . - (used to verify directory structure)

A terminal window with a dark background and a yellow 'M' icon in the title bar. The prompt is 'jmqque@DESKTOP-0ISP8AQ UCRT64 ~'. The command '\$ find .' has been entered, and the output lists the directory structure of the current directory. The files and directories listed are: ., ./.bashrc, ./.bash_history, ./.bash_profile, ./profile, ./hello.c, ./hello.exe, ./HelloWorld, ./KeyboardMouseEmulator, ./KeyboardMouseEmulator/emulator.exe, ./KeyboardMouseEmulator/main.c, ./OS, ./OS/Network, ./OS/Network/P2, ./OS/Network/P2/ReadMe1.txt, ./OS/Network/P3, ./OS/Network/P3/ReadMe2.txt, ./OS/P1, ./OS/P4, ./OS/P4/ReadMe1.txt, ./OS/P4/ReadMe2.txt, ./Script, ./Script/P1, ./Script/P1/ReadMe1.txt, ./Script/P3, and ./script101.sh.

```
jmqque@DESKTOP-0ISP8AQ UCRT64 ~  
$ find .  
.  
./bashrc  
./bash_history  
./bash_profile  
./profile  
./hello.c  
./hello.exe  
./HelloWorld  
./KeyboardMouseEmulator  
./KeyboardMouseEmulator/emulator.exe  
./KeyboardMouseEmulator/main.c  
./OS  
./OS/Network  
./OS/Network/P2  
./OS/Network/P2/ReadMe1.txt  
./OS/Network/P3  
./OS/Network/P3/ReadMe2.txt  
./OS/P1  
./OS/P4  
./OS/P4/ReadMe1.txt  
./OS/P4/ReadMe2.txt  
./Script  
./Script/P1  
./Script/P1/ReadMe1.txt  
./Script/P3  
./script101.sh
```

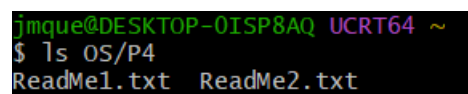
Step 9. Verify the Files

cat OS/P4/ReadMe1.txt

A terminal window showing the command '\$ cat OS/P4/ReadMe1.txt' entered at the prompt 'jmqque@DESKTOP-0ISP8AQ UCRT64 ~'.

```
jmqque@DESKTOP-0ISP8AQ UCRT64 ~  
$ cat OS/P4/ReadMe1.txt
```

ls OS/P4

A terminal window showing the command '\$ ls OS/P4' entered at the prompt 'jmqque@DESKTOP-0ISP8AQ UCRT64 ~'. The output is 'ReadMe1.txt ReadMe2.txt'.

```
jmqque@DESKTOP-0ISP8AQ UCRT64 ~  
$ ls OS/P4  
ReadMe1.txt  ReadMe2.txt
```